

Generation Interconnection System Impact Study Report

For Queue Project Q15-0515

New Hope Ap 345 kV

September 26, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0515. The Interconnection Customer proposes a 327.2 MW natural gas combined cycle facility interconnecting at the New Hope Ap 345 kV substation in the West Virginia area. The request was received on September 04, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 188 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,164,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HAYSI – TOM WATKINS 138 kV	CATAWBA – ATOKA 345 kV	114.1	266	13.2
TOM WATKINS – KIMPER 138 kV	ATOKA – CATAWBA 345 kV	104.1	305	9.8
SPRIGG SUBSTATION – KIMPER 138 kV	NEPTUNE – RAGLAND 345 kV L	102.1	243	7.1
CINDER – SPRIGG 138 kV	NEPTUNE – RAGLAND 345 kV	101.7	237	7.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HAYSI – TOM WATKINS 138 kV	Rebuild line with higher-capacity conductor	\$38,579,000	9% (\$3,458,000)
TOM WATKINS – KIMPER 138 kV	Reconductor line	\$2,775,000	8% (\$220,000)
SPRIGG SUBSTATION – KIMPER 138 kV	Reconductor line	\$4,737,000	8% (\$370,000)
CINDER – SPRIGG 138 kV	Reconductor line	\$1,486,000	8% (\$116,000)
	Total allocated to project		\$4,164,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.